CYP2C19 Variant Interpretation: Frequency Filtering, Prediction Coverage, Discordance and Limitations

Module: BIOT6900 — Module 5, Part 3

Gene: CYP2C19

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1. Thresholds and Calibration

1.1 Allele-frequency threshold

The analysis utilized a maximum allele-frequency threshold (MAX_AF) of 0.35. In the dataset, CYP2C19*2 (rs4244285) exhibited a global allele frequency of 0.168 (16.8%), while CYP2C19*17 (rs12248560) exhibited a frequency of 0.200 (20.0%) in gnomAD (Chen et al., 2024). Examining population-specific maxima in gnomAD shows that CYP2C19*2 reaches up to $\sim$ 29% in East Asian populations, and CYP2C19*17 reaches up to $\sim$ 20--25% in European and African populations. The selected MAX_AF threshold of 0.35 safely accommodates these common functional alleles while successfully removing two common benign variants, rs17885098 ($AF = 0.919$) and rs3758581 ($AF = 0.952$), both annotated as Benign in ClinVar with allele frequencies exceeding 0.9.

A stricter rare-disease threshold of 0.001 would have inappropriately removed 19 variants, including major pharmacogenetic alleles such as CYP2C19*2, CYP2C19*17, CYP2C19*3 (rs4986893), and CYP2C19*4 (rs28399504), alongside 14 other curated drug-response variants.

Interpretation: Pharmacogenomic alleles differ fundamentally from rare Mendelian disease variants because pharmacogenes operate under weak purifying selection, as phenotypic consequences typically manifest only upon exposure to specific xenobiotic medications rather than during embryonic development. Allele-frequency thresholds designed for Mendelian disorders (typically ≤ 0.001) assume that disease-causing mutations are extremely rare. In contrast, pharmacogenes frequently harbor common functional variants that were historically neutral or adaptive regarding xenobiotic metabolism. Applying a strict rare-disease filter purges the exact variants responsible for clinical inter-individual drug response variability. Furthermore, the two variants removed at the 0.35 threshold (rs17885098 and rs3758581) have frequencies > 0.9 simply because the human reference genome (GRCh38) carries the minor allele at those specific loci. Finally, retaining a variant below the frequency threshold only establishes that it is not excessively common in reference populations; it is **not** equivalent to classifying the variant as functionally important or pathogenic.

1.2 Calibration limitations

Calibration was constrained by the available benchmark data, comprising zero Pathogenic/Likely Pathogenic (P/LP) variants and five Benign/Likely Benign (B/LB) variants within the dataset. Notably, CYP2C19*17 (rs12248560) is included among these five calibration B/LB variants, annotated as “Benign; other” with a CADD score of 3.031. Including CYP2C19*17—a well-characterized, increased-function promoter allele—in the benign reference set helps define the baseline score behavior for functional non-damaging variants. However, the overall dataset lacked a balanced set of known pathogenic outcomes from which to assess how effectively prediction scores distinguish damaging from benign variants.

The calibration procedure returned a positive likelihood ratio (LR+) of zero for the scored band, described as “strong, benign.” This result must be interpreted in the context of the limited calibration sample. With zero P/LP examples, the analysis could not establish how the score behaves for known pathogenic variants. Therefore, the LR+ output is not reliable evidence that a variant is benign.

All 33 variants of uncertain significance (VUS) remained unresolved: 27 were uncalibrated due to missing P/LP labels, and 6 were unscored. These results demonstrate the distinction between a benign prediction supported by evidence and a variant that cannot be resolved using the available data.

Interpretation: The lack of positive benchmark examples affects the reliability of the calibration procedure because an empirical LR+ requires observations of true positives across score bins; with zero positive outcomes, the numerator collapses to zero regardless of score magnitude. Treating the resulting LR+ of zero as evidence of benign status would mistake a data gap for biological evidence. Similarly, the 33 unresolved VUS cannot be treated as benign; an uncalibrated status or missing score indicates an absence of computable evidence rather than proof of harmlessness.

2. Prediction Coverage and AlphaGenome Atlas

2.1 Coverage across prediction tools

Evaluation of tool coverage across the pre-filtered dataset ($n = 80$) revealed notable discrepancies among computational algorithms. REVEL and AlphaMissense scored missense variants (plus one start_lost variant), achieving a coverage of 67.5% (54/80). CADD covered 85.0% (68/80). The AlphaGenome Atlas returned scores for 62 of the 78 variants evaluated post-filtering.

Tool	Observed Coverage	Scope
REVEL	67.5% (54/80)	Missense and start-loss variants only (Ioannidis et al., 2016)
AlphaMissense	67.5% (54/80)	Missense variants only (Cheng et al., 2023)
CADD	85.0% (68/80)	Genome-wide (Rentzsch et al., 2019)
AlphaGenome Atlas	62/78 variants	Genome-wide regulatory and coding (DeepMind, 2024)

Interpretation: REVEL returned zero scores for the following consequence classes: intergenic_region, intron_variant, splice_donor_variant, stop_gained, stop_lost, synonymous_variant, upstream_gene_variant, and missing consequence annotations (NaN). A missing score from REVEL or AlphaMissense signifies that the variant class falls outside the tool's scope, rather than indicating that the tool evaluated the variant and found no functional effect. Furthermore, exactly eight variants in the dataset possessed no consequence annotation and no computational score (cadd_phred, revel, and alphamissense all NaN). Inspecting these entries reveals that they represent variant records lacking transcript consequence annotations in the underlying database export rather than complex structural intervals.

2.2 Atlas feature attribution

Among Atlas results, 34 variants were attributed to AlphaMissense, 17 to conservation (13 CACTUS_241_WAY and 4 PHASTCONS_470_WAY), 7 to splicing (MERGED_SPLICING), and 2 upstream variants to DNase-related features (MAX_ABS_DNASE). Conservation-based attributions highlight sequence constraint without resolving the underlying biological mechanism.

CYP2C19*17 (rs12248560) received an AVI score of 0.53 with a conservation attribution (CACTUS_241_WAY). An AVI Phred score of 0.53 is essentially negligible, ranking below most variants genome-wide. The top_feature attribute merely indicates which input contributed most to a near-zero score. Rather than indicating a functional signal, this result demonstrates that the Atlas did not flag *17 at all, even though *17 is an established increased-function promoter allele. This highlights that model insensitivity can miss established regulatory variants.

Another notable example is rs72558186 (CYP2C19*7), a splice-donor variant that lacked CADD, REVEL, and AlphaMissense scores. Its Atlas attribution was MERGED_SPLICING. Classifying rs72558186 requires experimental RNA assays, such as a minigene splicing assay or patient cDNA sequencing, to detect aberrant transcript processing.

Interpretation: A splicing attribution provides a testable hypothesis for further investigation rather than experimental proof of abnormal splicing. Furthermore, the Atlas did not score 16 of the 78 variants due to parsing constraints; these missing results should not be interpreted as benign predictions.

3. Discordant Variant: CYP2C19*2 (rs4244285)

CYP2C19*2 (rs4244285) was selected for further investigation because it triggered the common_but_plp and atlas_only discordance rules. Its observed allele frequency was 0.168, its consequence annotation was synonymous_variant, and its CADD PHRED score was 5.686. REVEL and AlphaMissense did not provide scores. In contrast, the AlphaGenome Atlas returned an AVI score of 22.33 and attributed the result to splicing (MERGED_SPLICING), and the dataset recorded a drug-response classification.

These results illustrate why variant annotation and computational prediction scores need to be interpreted in the context of molecular biology. A synonymous annotation describes the predicted coding consequence; it does not exclude effects on RNA processing. Similarly, a missing REVEL or AlphaMissense score cannot be treated as a benign prediction.

3.1 Mechanistic interpretation

CYP2C19*2 (c.681G>A, p.Pro227Pro) is a useful example of how a nucleotide substitution can have consequences that are not apparent from its immediate coding annotation. The established mechanism associated with CYP2C19*2 involves the creation of a cryptic splice site *within* exon 5, and the resulting mRNA lacks the first 40 bp of exon 5. That shifts the reading frame and produces a premature stop codon, leading to a truncated, non-functional protein (de Morais et al., 1994).

CADD is not a protein-level model; it is a genome-wide score integrating multiple annotations where conservation and constraint dominate. A synonymous change at a position under weak constraint, in a gene under weak purifying selection, looks unremarkable to a genome-wide constraint score (CADD PHRED 5.686). The AlphaGenome Atlas includes an explicit splicing component, which is where its strong signal originated, as reflected by the MERGED_SPLICING feature attribution.

Evaluating the alternative explanation—whether a linked variant on the haplotype contributes to the observed association—is addressed by noting that the Atlas attribution pins the splicing disruption signal directly to this coordinate. However, confirming whether this single G>A change is sufficient requires empirical testing.

Connection to Part 1: The Part 1 example, g.39C>T, provides a comparison for considering how a nucleotide substitution can have downstream consequences that are not obvious from its immediate annotation. Specifically, g.39C>T was a synonymous variant that created a cryptic splice donor (donor_gain 1.000, at the same strength as the real site). That makes it an almost exact parallel to *2, strengthening the mechanistic hypothesis.

Overall, discordance is a reason to investigate the molecular mechanism rather than treating any one computational score as a clinical verdict.

4. Limitations and Proposed Next Experiment

4.1 Zygosity and phase

The analysis describes individual variants but does not establish the complete genotype of a particular patient. It therefore cannot determine whether an individual carries one or two copies of *2 or whether multiple variants occur on the same chromosome.

Interpretation: Two reduced-function variants in cis sit on the same chromosome, leaving the other chromosome functional; this yields roughly an intermediate metabolizer phenotype rather than normal activity. In trans, both chromosomes are impaired, resulting in a poor metabolizer phenotype. Variant-level tables cannot resolve phase.

4.2 Clinical utility

A computational prediction of molecular impact does not establish whether a genotype-guided treatment decision improves outcomes for a patient. Clinical effects depend on prescribed medications, co-medications, and individual characteristics.

Interpretation: Determining clinical utility requires prospective clinical implementation trials (such as the PREPARE trial; Swen et al., 2023) demonstrating that pre-emptive panel testing reduces adverse events.

4.3 Proposed experiment

A minigene splicing assay testing the single c.681G>A construct (isolated from the rest of the haplotype) could test whether the alternate sequence changes splicing compared to the reference sequence.

- **Expected result:** The alternate construct will produce an aberrant transcript lacking the first 40 bp of exon 5 due to cryptic splice site activation, whereas the reference construct will produce normally spliced mRNA.
- **Result that would change my interpretation:** Observing no reproducible difference in splicing between reference and alternate constructs would indicate that c.681G>A alone is insufficient to disrupt splicing, reducing confidence in the isolated variant hypothesis.

5. Technical Disclosures and Reproducibility

- **Environment and Execution:** The notebook ran in the base Anaconda environment. AlphaGenome was installed using `%pip install "alphagenome>=0.9.0"`.
- **Tool Coverage & Atlas Status:** The Atlas returned scores for 62 of 78 variants. Sixteen variants were not scored. Missing scores were treated as limitations of coverage.
- **Merging and Data Handling:** Left joins were used when merging Atlas outputs. Two distinct data handling issues occurred:
    1. The empty `top_feature` column resulted from rerunning the Atlas cell after an initial interruption, which was resolved by reloading from the saved file.
    2. A duplicate-rsid issue occurred because rs4244285 possesses two alternate allele rows, causing the second row to inherit a copied AVI score upon merging. This duplicate row retaining a copied AVI is disclosed as a technical limitation.
- **Calibration:** Calibration was reported as impossible given the available labelled variants.

References

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